

TUMBES, Peru

WMO: 843700

Lat: 3.553S

Lon: 80.381W

Elev: 35

StdP: 100.90

Time Zone: -5.00 (W05)

Period: 82-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.3	19.9	15.7	11.2	22.0	16.8	12.1	21.5	8.3	25.4	7.3	25.8	0.8	270	0.260

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.6	32.1	25.7	31.7	25.4	31.1	25.1	26.6	30.5	26.2	30.0	25.9	29.7	4.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.1	28.7	25.2	20.4	28.2	24.9	20.0	28.0	83.6	30.9	81.8	29.9	80.5	30.0	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.7	5.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	15.1	27.7	2.6	1.6	13.2	28.8	11.7	29.7	10.2	30.6	8.3	31.7	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	26.8	27.4	27.6	27.1	26.6	25.1	24.3	23.7	23.8	24.6	25.1	26.2
	DBStd	1.98	1.02	0.91	0.95	0.99	1.29	1.75	1.82	1.89	1.48	1.62	1.61	1.32
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5721	521	487	544	514	514	453	442	424	413	451	452	504
	CDD18.3	2679	263	253	286	264	256	203	184	166	163	193	202	246
	CDH23.3	20070	2272	2477	2858	2447	2141	1302	979	758	672	1007	1223	1934
(13)	CDH26.7	5269	614	753	911	748	587	305	202	156	94	183	242	475
(14)	Wind	WSAvg	2.0	2.1	2.0	2.2	1.8	1.7	1.9	2.0	2.1	2.2	2.1	2.2
(15) Precipitation	PrecAvg	482	59	137	157	77	15	1	1	1	3	5	7	21
	PrecMax	2238	438	734	508	426	140	12	11	5	25	19	140	429
	PrecMin	41	0	0	17	0	0	0	0	0	0	1	0	0
	PrecStd	436	86	159	140	90	35	3	2	1	5	4	26	81
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.6	32.8	32.4	32.2	31.8	31.6	31.5	30.7	31.1	31.8	32.0
		MCWB	25.2	25.9	26.5	25.9	25.7	25.5	25.2	24.9	24.6	24.6	24.9	25.2
	2%	DB	31.0	31.8	32.0	31.7	31.2	30.6	30.3	30.1	29.0	29.9	30.2	30.9
		MCWB	24.7	25.5	25.8	25.7	25.3	25.1	24.9	24.4	23.7	24.0	24.4	24.9
	5%	DB	30.2	31.0	31.2	31.0	30.4	29.6	28.8	28.2	27.5	28.7	29.0	30.0
		MCWB	24.3	24.9	25.3	25.3	24.8	24.5	24.1	23.5	22.8	23.1	23.4	24.3
	10%	DB	29.8	30.2	30.5	30.2	29.7	28.3	27.3	26.6	26.3	27.4	28.0	29.1
		MCWB	24.1	24.6	25.0	24.8	24.4	23.8	23.0	22.6	22.2	22.7	23.0	23.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	27.2	27.2	26.9	26.7	26.2	26.0	25.2	25.0	25.1	25.6	25.9
		MCDB	29.8	30.8	31.5	31.2	30.4	30.0	30.1	30.5	30.0	30.4	30.8	30.5
	2%	WB	26.0	26.3	26.5	26.2	26.1	25.6	25.2	24.6	24.1	24.3	24.7	25.3
		MCDB	29.4	30.1	30.4	30.4	29.7	29.3	29.5	29.6	28.2	29.0	29.5	29.7
	5%	WB	25.4	25.9	26.1	25.8	25.6	25.0	24.4	23.7	23.3	23.6	24.1	24.9
		MCDB	28.7	29.8	29.7	30.0	29.2	28.8	28.3	27.8	26.8	27.8	28.1	29.1
	10%	WB	24.9	25.4	25.6	25.3	25.0	24.3	23.6	22.9	22.6	23.0	23.4	24.4
		MCDB	28.1	29.0	29.1	29.1	28.6	27.7	26.8	26.2	25.8	27.0	27.2	28.2
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.6	6.6	6.8	6.6	6.1	6.0	5.8	5.4	5.6	5.8	6.1
		MCDBR	7.0	7.3	7.3	7.7	7.4	7.4	7.5	7.9	6.7	6.8	6.9	6.8
		MCWBR	3.1	3.0	3.0	3.1	3.1	3.3	3.7	3.8	3.3	3.2	3.3	3.1
	5% WB	MCDBR	6.5	7.0	6.9	7.5	7.2	7.3	7.5	7.6	6.6	6.4	6.5	6.6
		MCWBR	3.1	3.3	3.3	3.2	3.3	3.3	3.8	3.9	3.4	3.1	3.5	3.1
(40) Clear-Sky Solar Irradiance	taub		0.466	0.458	0.469	0.484	0.479	0.457	0.461	0.478	0.523	0.502	0.496	0.505
	taud		2.315	2.347	2.313	2.260	2.275	2.347	2.334	2.270	2.123	2.195	2.200	2.176
	Ebn at Noon		873	881	861	825	803	808	811	819	804	833	840	833
	Edh at Noon		136	133	136	139	131	120	123	136	162	153	152	155
(44) All-Sky Solar Radiation	RadAvg		5.05	4.97	5.30	5.36	4.86	4.43	4.38	4.67	5.03	5.20	5.39	5.32
	RadStd		0.41	0.28	0.34	0.21	0.27	0.27	0.32	0.30	0.31	0.25	0.29	0.30

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page